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Gabe Ignatow & Laura Robinson

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Pierre Bourdieu: theorizing the digital

Gabe Ignatow ^a and Laura Robinson^b

^aDepartment of Sociology, University of North Texas, Denton, TX, USA; ^bDepartment of Sociology, Santa Clara University, Santa Clara, CA, USA

ABSTRACT

Pierre Bourdieu is known for his research in the areas of education and cultural stratification that led to a number of theoretical contributions informing the social sciences. Bourdieu's interrelated concepts of field, capital, and habitus have become central in many approaches to inequality and stratification across the social sciences. In addition, we argue that Bourdieu's ideas also feature in what is increasingly known as 'digital sociology.' To underscore this claim, we explore the ways in which Bourdieu's ideas continue to have a major impact on social science research both on and with digital and Internet-based technologies. To do so, we offer a review of both Bourdieusian theorizing of the digital vis-à-vis both research on the social impacts of digital communication technologies and the application of digital technologies to social science research methods. We contend that three interconnected features of Bourdieu's sociology have allowed his approach to flourish in the digital age: (1) his theories' inseparability from the practice of empirical research; (2) his ontological stance combining realism and social constructionism; and (3) his familiarity with concepts developed in other disciplines and participation in interdisciplinary collaborative projects. We not only reason that these three factors go some way in accounting for Bourdieu's influence in many sociological subfields, but we also suggest that they have been especially successful in positioning Bourdieusian sociology to take advantage of opportunities associated with digital communication technologies.

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Introduction

Pierre Bourdieu is widely considered to be among the most influential late-twentieth-century social theorists, with theoretical and empirical contributions ranging over fields such as political sociology, education, and cultural stratification. His influence on social science research is evident not only in these areas, which he studied intensively and wrote about extensively, but also in at least one area which he never touched upon at all, namely, digital communication technologies. Over the past 15–20 years, even before his death in 2002, Bourdieu's development of the interrelated concepts of field, capital, and habitus has informed what is increasingly termed 'digital sociology' (Daniels, Gregory, & Cottom, 2016; Lupton, 2014; Marres, 2017; Orton-Johnson & Prior, 2013). The term 'digital sociology' can refer

CONTACT Gabe Ignatow  ignatow@unt.edu  Department of Sociology, University of North Texas, 1155 Union Circle #311157, Denton, TX 76203, USA

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both to research on the social aspects and impacts of digital communication technologies and to the application of digital technologies to research methodologies across the social sciences.

Bourdieu developed his approach to social fields and capital in publications such as *Distinction* (1984) and *An invitation to reflexive sociology* (Bourdieu & Wacquant, 1992), and to *habitus* in several publications beginning in the 1960s (Bourdieu, 1966, 1971, 1980). Bourdieu sought to move social science away from variable-centered hypothesis-testing research toward a relational approach to the study of social life. This approach conceptualizes social action as occurring within a social space made up of intersecting fields conditioning and constraining the behavior of individuals and shaping their internal motivational apparatus. For Bourdieu, ‘the real is the relational’ (Bourdieu & Wacquant, 1992, p. 97) such that what might otherwise appear to be disparate categories of social phenomena, such as social structures and mental structures, or students’ aesthetic preferences and academic performance, are interrelated and entangled. Bourdieu’s main concepts are not only defined relative to this overarching ontology, but they are also interrelated in such a way that they can only make sense in relation to each other.

In what follows, we assume some background knowledge of Bourdieusian social science on the part of our readers (Swartz, 1997). We survey the substantive and methodological terrain of digital sociology (and related social sciences) that draws on or has elective affinities with Bourdieu’s oeuvre, and draw lessons from this survey for future sociological research. Specifically, we suggest that there are substantive reasons why the Bourdieusian research program has gained traction within the field of digital sociology, under social circumstances which he could not possibly have anticipated when he was devising his theory. Drawing on the secondary literature on Bourdieu, we contend that three interconnected features of Bourdieu’s approach have enabled his approach to flourish even as other social and sociological theories have struggled to demonstrate their relevance in the digital age: (1) his theories’ inseparability from the practice of empirical research; (2) his ontological stance combining realism and social constructionism; and (3) his familiarity with concepts developed in other disciplines and participation in interdisciplinary collaborative projects. These three factors go some way in accounting for Bourdieu’s influence in many sociological subfields, but we suggest that they have been especially successful in positioning the Bourdieusian approach to take advantage of opportunities associated with digital communication technologies.

Pierre Bourdieu: theorizing the digital

Numerous studies apply established research approaches and methods to the study of social phenomena mediated by social media platforms and other digital communication technologies. Many of these studies draw heavily on the Bourdieusian approach. As this body of research is so voluminous, we turn our attention to one of the subfields of digital sociology that makes significant use of Bourdieu’s concepts: the study of digital inequality. We review influential studies organized in terms of Bourdieu’s signature concepts of field, capital, and habitus in order to explore the continuing impact of these concepts.

Field theory

‘Field’ (*champ*) is a key spatial metaphor for Bourdieu that differentiates his work. Field theory represents an ‘interest in forces, intensities, dynamics and processes, in place of

a more static sociology of variables, categories and social groups' (Savage & Silva, 2013, p. 111). Bourdieu defines a field as a network or configuration of relations between social positions in which positions and their interrelations are determined by the distribution of economic, social, and cultural capital. Though the borders between fields are porous, each field is characterized by its own logic (the 'rules of the game'). Actors within fields struggle to accumulate and monopolize capital based on the field-specific rules of the game, with more successful actors being more adept at both accumulating and reinvesting capital (Bourdieu, 1994; Bourdieu & Wacquant, 1992).

Bourdieu's treatment of the concept of a social field is far from the only contemporary treatment (Fligstein, 2001; Fligstein & McAdam, 2012; Martin, 2003). Nevertheless, as the field of digital inequality research has grown, researchers are increasingly using Bourdieusian field theory as a foundation for their work (Hilgers & Mangez, 2015). For instance, the field concept has been used in survey research on status differences in Internet usage. Exemplary research in this area includes the work of Austrian sociologist Zillien and her colleagues on the digital divide in Europe (Zillien & Marr, 2013), work by Hargittai and her colleagues on Internet access and use patterns in the U.S.A. (Hargittai & Hinnant, 2008), and Arie and Mesch's research on digital inequality in Israel (2015). Levina and Arriaga (2014) introduce the notion of an 'online field' as an analytical lens for studying social status production processes on User-Generated Content platforms. Their goal is to explain how diverse types of producers and consumers of content jointly generate unique power relations online and what role platform design choices play in shaping which forms of distinction count and how they are pursued.

Information capital

The concept of capital is inseparably linked to the concept of field. For Bourdieu, capital refers to stocks of internalized ability and aptitude as well as externalized resources which are scarce and socially valued. Like the more traditional form of capital, they can be transformed and productively reinvested. For instance, one of the primary forms of capital, economic capital, in the form of money can be exchanged for cultural capital in the form of higher education, which in turn may facilitate the accumulation of more economic capital over the life course as well as social capital in the form of relationships with high-capital classmates and instructors. For Bourdieu, actors' positions within various social fields correspond with the volumes of the different forms of capital they possess. Capital has come to be a centrally important concept in studies of digital inequality, with sociologists developing and employing in empirical research concepts such as 'information capital' and 'digital capital.' Although the earliest concept of information capital was developed by Hamelink (2000), it is van Dijk's (2005) comprehensive definition that has had the greatest influence. Van Dijk defines information capital as the financial resources to pay for computers and networks, technical skills, evaluation abilities, information-seeking motivation, and the capacity for implementation (2005, pp. 72–73). An alternative approach more faithful to the Bourdieusian program is to define digital capital as a secondary form of capital distinct from primary forms of capital such as economic and cultural capital. In this view, a person's stock of digital capital corresponds to the reach, scale, and sophistication of his or her online behavior. It is important to note here that there are particular forms of digital capital which are readily convertible into economic capital, such

as programming ability, whereas other kinds of digital capital, such as social media activity, can be converted into social capital, but do not typically make the holder more attractive on the labor market.

The Bourdieusian framework and the study of digital inequality

The Bourdieusian framework has proved useful in empirical studies which explore the nature of digital ability across different kinds of groups and life activities and the links between digital capital and non-digital forms of capital. Such studies have shown how the same kinds of informational engagements yield different payoffs for more and less disadvantaged groups. Witte and Mannon (2010) use representative survey data to show that inequalities in IT access and Internet usage both augment and mirror inequalities in offline resources such as economic and cultural capital. Kvasny (2005) examines how perceptions of IT differ according to socio-demographic background. Drawing on the concept of information capital, other studies have also indicated the importance of acquiring particular 'digital skills' (van Dijk, 2005, p. 73). Skills related to finding and assessing information constitute one of the building blocks of information literacy. Mastery of digital skills is a precondition for the acquisition of informational advantage. Not only do more-skilled Internet users reap benefits by obtaining desired information with less effort, but they also use the Internet in a more flexible and versatile manner than less-skilled users. Studies have found that more-skilled users transition more easily from one website to another and enlist the Internet for a more varied menu of human capital-enhancing activities (Witte & Mannon, 2010, pp. 95–113). Skills allow users to use the Internet effectively, which in turn gives 'wired' individuals advantage compared to their less-wired counterparts in personal and professional life spheres (DiMaggio & Bonikowski, 2008).

Several recent studies underline the importance of capital-enhancing digital activities and skills regardless of the national context. Three such studies reveal a similar pattern in the U.S.A., Denmark, and Peru. Drawing on data from the U.S.A., McConnell and Straubhaar examine the use of open WiFi networks in the City of Austin, Texas. Building on Bourdieu's concept of multiple forms of capital, they find that both educational capital and 'technocapital' predict greater usage of open WiFi systems. At the same time, they find that individuals without home access, who could benefit the most from open WiFi, do not make use of it to the same extent as those users also have greater forms of capital. Thus, they argue that 'simply offering internet services via wifi is likely ineffective in expanding internet use among disadvantaged populations.' van Deursen and Helsper (2015) demonstrate that more socioeconomically advantaged Internet users acquire greater digital capital and derive greater benefits from Internet usage, as contrasted with less advantaged counterparts in highly wired societies such as Denmark. This pattern also holds true in less wired nations such as Peru, according to Villanueva-Mansilla, Nakano, and Evaristo (2015) who demonstrate linkages between digital divides and social and cultural capital. Based on data from a private university in Peru, they find connections between self-perception of access and skills, differentiated media use, and divergent forms of digital capital. They distinguish between the capital that is 'spent and accrued in social relationships' and the digital productive capital that is 'spent and accrued on formal educational contexts.' They argue that, among these students, digital capital assumes disparate forms, depending on the users' activity patterns; while some users build up digital capital specific to social

media activities, others accrue digital capital particular to human capital enhancement, typically associated with educational achievement. Thus, they distinguish the form of digital capital implicated in the user-defined seeking of social gratifications from the form of digital capital implicated in institutionally structured human capital enhancement. This form of capital reflects objectives defined and mediated through institutions such as schools.

The *habitus* is the internalization of the field, a set of historical relations incorporated within individual bodies in the form of mental and corporeal schemata. These schemata are acquired in daily life through social interactions, and include schemata related to comportment (posture and gait), aesthetic likes and dislikes (taste), habitual linguistic practices (accent, vocabulary, the speed, and volume of spoken language), and ways of evaluating oneself and others via categories such as refined/vulgar and masculine/feminine. The *habitus* is the centerpiece of the 'third phase' of Bourdieu's reception (Lizardo 2012), a phase that has seen significant sociological interest (Ignatow, 2007). Using the concept of the *habitus* illuminates the importance of Bourdieu's work to the field of digital inequality. Importing a Bourdieusian framework into the digital realm allows us to grasp how individuals relate to IT resources, specifically how differently situated individuals' informational *habitus* emerges from long-term experiences of scarcity and abundance with respect to other primary goods.

For example, Robinson draws on a Bourdieusian framework to argue that disparities in the level of Internet skills originate in inequalities of access, but are mediated by orientations that can only be understood in relation to total life contexts. She refines the information *habitus* concept to capture how ways of interacting with digital technologies become habitualized by individuals operating within local social contexts and field positions. Robinson finds two types of information *habitus* in her ethnographic work on information communication technology (ICT) use among low- and middle-income families in an agricultural belt of California. The playful *habitus*, common within upper-middle-income families, is *skholè*, or 'serious play,' in which ICT use is encouraged; the latter *habitus* promotes a deep level of technological engagement that engenders skill development over the long term (Robinson, 2009). By contrast, disadvantaged youths develop a task-oriented information *habitus* in which they enact a Bourdieusian 'taste for the necessary.' They ration their Internet use in response to experiences of constraint and temporal urgency, which Bourdieu (1994) reminds us emerges from access to accumulated economic resources. At the mercy of spatial-temporal urgencies that do not encumber their more advantaged peers, these youth develop a task-oriented information *habitus* antithetical to *skholè* in which waste avoidance is their primary goal. By enacting a taste for the necessary, they do not acquire the same skills and benefits as their more advantaged peers who are playing seriously. Youths who do not receive emotional rewards for Internet use are less likely to use the Internet for education and career information searches later in life.

The enactment of a 'taste for the necessary' is ultimately counterproductive and reinforces disadvantage. An example is provided in another study by Robinson that builds on the notion of information *habitus* to understand how advantaged and disadvantaged youths acquire digital information central to their educational and career planning (2011). Here, Bourdieu's concept of the *habitus* is indispensable to revealing actions, decisions, and orientations that help to define life chances and futures. Deepening the

concept of the information habitus in this context reveals how advantaged and disadvantaged youth internalize different stances toward using digital technologies for their post-secondary education and career planning. Robinson finds that just as adolescents internalize stances toward the appropriate use of the Internet, they also internalize stances toward appropriate information gathering for vocational and educational planning based on what they believe to be the perceived costs and payoffs of each information channel. For youths enacting a task-oriented habitus, the Internet reinforces pre-existing social disparities because these students are less willing to use the Internet in their education and career information searches. Given the task-oriented information habitus that they have internalized, these students are far less likely to employ digital media in ways that could improve their life chances in one of the most influential life realms: career and college planning.

A number of studies of the digital habitus distinguish between a habitus focused on capital-enhancing activities versus one focused on recreational activities. Several studies reveal systematic differences in the weight given to these two kinds of uses by users from distinct class backgrounds. This is particularly true for education researchers who have examined digital literacy and families' use of ICTs either as recreational or as tools for 'concerted cultivation' (Lareau, 2011). In such work, upper-middle-class youths often participate in institution-conforming digital practices revolving around work or school and self-directed digital practices typically revolving around recreation, socializing, and leisure pursuits. For example, Micheli (2015) examines the mediation of Italian youths' digital activities by their social environments and backgrounds. She finds that whereas information-seeking positively correlates with students' cultural capital and parental occupational status, social media use does not. She finds that, contrary to expectations, teenagers from less advantageous social backgrounds enrolled in vocational schools have better chances to actively participate in social media than do teens from the upper-middle class in academically oriented high schools. Her interpretative analysis of the qualitative data indicates that upper-middle-class students attending *licei* replicate their parents' stance toward the Internet as a tool for personal enrichment. By contrast, teens attending vocational school engage with digital media as a form of peer-oriented leisure. Her findings are an exemplar of research in this vein, revealing that students from more elevated class backgrounds spend most of their time online seeking out information, a capital-enhancing activity directed by schools and educators. Users from lower and working-class backgrounds spend more of their online time engaging with social media sites and games, activities which have little to do with schooling or human capital enhancement. Like many studies using the information habitus concept, Micheli's work indicates the importance of the family. In families, the information habitus can create an environment priming academic achievement (Huang & Russell, 2006) or entertainment. Families that domesticate ICTs in favor of capital-enhancing activities will allow individuals more time to pursue them. By contrast, families that prefer to use ICTs for entertainment will assign greater weight to more pleasurable pursuits (Robinson & Schulz, 2013).

As this section has shown, Bourdieusian concepts of the field, capital, and habitus are at the heart of one of the key subfields in digital sociology: digital inequality. This is confirmed by a recent study by a number of digital inequality scholars (Robinson et al., 2015) who examine the significance of digital inequalities across a broad range of individual- and

macro-level domains, including life course (Cotten & Gupta, 2004), gender (Ono & Zavodny, 2007), race (Mesch, 2006), and class (Hale, Goldner, Stern, Drentea, & Cotten, 2014; Stern, Adams, & Elsasser, 2009). Robinson and her colleagues argue that digital inequality has implications in all major life spheres including healthcare, politics, economic activity, and social capital. Further, they call for new methods for digital sociology. In the next section, we address how researchers are answering this call.

Bourdieuian digital sociology

Bourdieuian concepts have proven effective when applied to substantive areas in digital sociology such as the study of digital inequality. These concepts have also been used by researchers developing social research methods based on data derived from the digital footprints left by individuals' activities in the online realm. For the sake of convenience, we review research that uses each of Bourdieu's major concepts separately.

Field theory

A number of social science studies have approached online social phenomena in terms of fields. While field analysis has been 'trumpeted as a fundamental means of renewing social scientific analysis' (Savage & Silva, 2013, p. 111), its appeal is not exclusively theoretical but 'fundamentally includes methodological repertoires. Such innovations are potentially more important and exciting' (Savage & Silva, 2013, p. 114). While social science methods are conventionally divided into 'variable-centered' quantitative methods and more qualitative 'case-centered' methods, field analysis requires different sorts of strategies to assess the relationships between different elements of fields.

Field theory forms the basis of several new digital research methods as well as novel applications of established methods. Bourdieu himself, in collaboration with numerous statisticians (Lebaron, 2009), used multiple correspondence analysis (MCA), most notably in *Distinction* (1984). Rather than seeking to explain causal relationships between independent and dependent variables one set at a time, correspondence analysis seeks to elaborate complex relationships between different phenomena. Correspondence analysis represents a set of cultural items in a dimensional space, allowing their underlying structural locations to be visually represented. In *Distinction* (1984), Bourdieu used data on the cultural tastes of different class fractions to identify the class-based logic of cultural consumption in France. Bourdieu and his collaborators used correspondence analysis to simultaneously plot various social locations according to their relative location within the space of cultural distinctions.

Anglophone cultural sociologists have used correspondence analysis to represent multiple relationships in data sets with complex structures. For instance, Mische (1998) used correspondence analysis to analyze the political discourses of youth movements in Brazil. Friedland, Mohr, Roose, and Gardinali (2014) used correspondence analysis for visualizing the relations between topics (Mohr & Bogdanov, 2013) from a survey of ideas about love given to American university students. And British sociologists Savage and Gayo-Cal (2011) used correspondence analysis in a comprehensive field analysis of the structure of British musical taste based on survey questions and qualitative interviews. Beyond MCA, there are many other recently developed digital visualization tools that are relational and

multidimensional in orientation rather than variable-centered. These include concept maps, mind maps, path and network diagrams, word clouds (Viégas & Wattenberg, 2008), word trees, phrase nets, and matrices that can be used for visually representing the multidimensional interrelations of social and linguistic phenomena in quantitative, qualitative, and mixed methods research (Wheeldon & Åhlberg, 2012).

Capital

Sociologists have developed several new survey tools to measure ‘digital capital.’ For example, Seale, Ziebland, and Charteris-Black (2006) have developed a ‘digital capital’ framework to explore the relationship between disabled students in higher education and their use of learning support technologies. Collecting data from disabled students in a teaching-intensive university in the U.K. using an online questionnaire survey and a follow-up semi-structured interview, the researchers found that while disabled students do have access to social and cultural resources, sometimes these resources are not appropriate or effective or they are not drawing on all the possible resources available to them.

Other studies have focused on digital dimensions of social capital by using data available from social media platforms such as Facebook and Twitter (e.g., Hofer & Aubert, 2013). For instance, based on their analysis of Facebook user data, Lewis, Kaufman, Gonzalez, Wimmer, and Christakis (2008) found that subgroups defined by gender, race/ethnicity, and socioeconomic status were characterized by distinct network behaviors, and that users sharing social relationships and demographic traits tended to share cultural preferences as well. Brooks, Hogan, Ellison, Lampe, and Vitak (2014) analyzed social network structures using Facebook. Administering a survey including measures of self-reported Facebook activity to 235 employees at a Midwestern American university, Brooks et al. analyzed differences in bonding social capital between social networks with different structural patterns.

Digital sociological studies of cultural capital are even more explicitly Bourdieusian than are studies of social capital. Paino and Renzulli (2013), for example, argue for broadening the definition of culturally important forms of capital to include the digital dimension of cultural capital. They suggest that students who possess knowledge of computers and other digital devices may gain actual skills, but more importantly, they are constituting and representing themselves as culturally competent members of our information-age society. Bourdieu’s conception of cultural capital would predict that those students who possess and exhibit it as measured by cultural activities, such as attendance at a museum or participation in dance, will be more likely to succeed educationally. Paino and Renzulli suggest that the same should hold true for the digital dimension of cultural capital. Based on survey data from primary school students, their main finding is that teachers play a prominent mediating role in the effects of computer proficiency on academic achievement. In another digital capital study, Hatlevik, Guðmundsdóttir, and Loi (2015) developed a ‘digital competence quiz’ to measure differences in secondary students’ digital competence.

Nissenbaum and Shifman (2015) have taken digital cultural capital research in a new direction. They explore the association between cultural capital and Internet memes (digital items that are imitated and reiterated around the Internet), arguing that the contradiction between following conventions and supplying innovative content leads to memes’

configuration as unstable equilibria. This triggers constant conflict about their ‘correct’ use. This contradiction and its resulting conflicts highlight collective identity by keeping shared culture at the center of discussion. Nissenbaum and Shifman analyze the website 4chan’s irreverent /b/ board by tracing keywords to investigate cases of social violations and condemnation. They find that seemingly trivial humor has social functions on 4chan and, more generally, that 4chan memes are unstable cultural forms whose instability contributes to community cohesion.

Habitus

Several digital sociological research methods have been developed that quantify aspects of habitus. In order to analyze digital content, these methods approach language as a product of mind–body interactions. For example, researchers have developed methods for coding sentiment as expressed in texts, including methods of human coding that rely on repeated readings of texts and the development of codes for sentiment words. Human coding of sentiment in large text collections has the advantage of capturing nuances in emotional expression that are often missed by highly automated methods. Human coding can be partially automated through the application of supervised learning techniques (Grimmer & Stewart, 2013, pp. 9–10). Another approach to sentiment analysis involves using dictionary-based sentiment analysis tools. Dictionary-based methods do not rely on ad hoc human coding of texts, but instead use off-the-shelf, validated sentiment lexicons. Dictionary methods involve using the rate at which keywords appear in a text to classify documents into categories or to measure the extent to which documents belong to particular categories. They measure a document’s tone by using a list of words with attached tone scores and the relative rate at which these words occur. A dictionary to measure tone is simply a list of words that are classified as positive or negative, either dichotomously or continuously. For example, dictionaries have been used by political scientists to measure the tone of newspaper articles (Eshbaugh-Soha, 2010) and political texts such as speeches and advertisements (Young & Soroka, 2011).

A second digital sociological method for analyzing the habitus is the computational analysis of metaphorical language. Metaphorical language provides striking evidence of how language is embodied (Lakoff & Johnson, 1980). Metaphors are highly parsimonious form–meaning pairings because they express meaning by enacting bodily and emotional operations. Ignatow (2003, 2004, 2007) and several other sociologists (Schmidt, 2012; Schuster, Beune, & Stronks, 2011) have developed sociologically oriented methods of metaphor analysis, but have thus far stopped short of developing highly automated methods that take advantage of currently available digital technologies and data. Today several research teams in computational linguistics and related fields are developing methods for automatically detecting metaphors in texts. For example, Neuman et al. (2013) have developed a number of interrelated algorithms that have proven highly accurate in identifying figurative versus non-figurative language.

Explaining Bourdieu’s impact on digital social science

Bourdieu’s concepts have become central to digital sociology both substantively and methodologically. What did Bourdieu get right that allowed his approach to thrive in foreign

terrain and in socio-technical circumstances he could not have anticipated? Why are digital sociological methods evolving in ways that reflect Bourdieu's understanding of the interactions of social inequalities and cultural and physiological phenomena? We suggest that these are questions worthy of extended reflection and discussion by sociologists interested in sociology's future development in the digital age. We further suggest that the success of Bourdieu's conceptual contributions in such unanticipated circumstances is no mere accident of historical timing or the popularity of his approach among social scientists. Drawing on the large secondary literature on Bourdieu as well as on contemporary philosophy of social science, we contend that three interrelated features of Bourdieu's sociology have allowed his approach to flourish even as other social and sociological theories have struggled for relevance in the digital age. These are the following: (1) his theories' inseparability from empirical research; (2) his ontological stance combining elements of realism and constructionism; and (3) his familiarity with concepts developed in other disciplines and participation in interdisciplinary collaborative research projects. These features go some way toward accounting for Bourdieu's influence in sociology. They have been especially successful in putting Bourdieusian sociology in a position to capitalize on research opportunities afforded by digital communication technologies. In arguing for the importance of these three interdependent factors, we do not promote any particular strand of contemporary Bourdieusian sociology, but rather attempt to recast how we understand Bourdieu's legacy and its relevance to the continuing development of sociology as it attempts to contribute to our understanding of contemporary information societies.

Grounding in empirical research

Though Bourdieu wrote as a theorist, he nevertheless 'sharply criticizes "theoretical theory" for emphasizing abstract conceptualization independent of objects of empirical investigation' (Swartz, 1997, p. 5). In his empirical research Bourdieu studied, among other topics, colonial Algeria, higher education, and class-based lifestyles and consumption habits. The global influence of his approach stems in large measure from the ease with which his theoretical apparatus accommodates the practical needs of empirical social science. Surveying Bourdieu's influence on American sociology, for example, Sallaz and Zavisca (2007) find that Bourdieu 'has been actively put to use to generate new empirical research' and has inspired a 'progressive research program' in the four core sociological subfields of political, economic, cultural, and urban sociology. Sallaz and Zavisca's quantitative analysis of citation patterns and case studies of books reveals that 49% of the publications in their sample used Bourdieu's ideas.

Bourdieu's writings have informed empirical research across many subfields, including the sociology of ethnicity and nationalism (Brubaker, 2004), media studies (Benson & Neveu, 2005), education (Carter, 2005), and the family (Lareau 2011). In political sociology, Sallaz and Zavisca discuss both Ron's (2000) work on Israeli soldiers who were interviewed regarding the use of repression during actions against Palestinians, and Eyal, Szelenyi, and Townsley's (1998) study of the emergence of capitalism in postcommunist Central Europe. In economic sociology, Sallaz and Zavisca give the example of Fligstein's *The architecture of markets* (2002) which used the field concept to illuminate why countries differ in their dominant employment systems, the evolution of corporate

management styles in the U.S.A., and the dynamics of globalization. While Bourdieu-influenced cultural sociology consists mostly of survey research on the association between cultural capital and highbrow taste (Holt, 1997), Sallaz and Zavisca note that Lamont's *Money, morals and manners* (1994) took Bourdieu's ideas in new directions in analyzing how French and American upper-middle-class White men draw 'symbolic boundaries' to define themselves and classify others. A follow-up book, *The dignity of working men* (2000), extended Lamont's study to working-class and nonwhite men in both countries. In urban sociology, Sallaz and Zavisca give the example of Wacquant's (2004) ethnographic exploration of habitus in *Body and soul*. Wacquant used participant observation data collected while he was a graduate student at the University of Chicago to describe the genesis and functioning of the pugilistic habitus in a boxing gym situated in the city's impoverished south side. In digital sociology, as we have shown, Bourdieu's ideas have been influential in both the empirical analysis of digital inequality and the development of new empirical research methodologies.

Philosophical foundations

Beginning in the 1970s, Bourdieu developed ontological and epistemological positions as the basis for his empirical sociological research. The positions he established prefigured a number of contemporary philosophy of social science approaches such as 'neo-materialism, post-positivist realism, critical realism, and critical sociomaterialism,' which have in common 'a sense of scientific practice as value-laden and productive of the objects it studies, combined with an interest in and acceptance of the capacities of matter beyond or before human interpretation' (Pitts-Taylor, 2014).

Like critical realist philosophy of social science (Bhaskar, 2013), Bourdieu's ontology integrated elements of social constructionism and realism. But in critical respects Bourdieu's approach more closely resembles (and is more frequently cited by) second-generation critical realist philosophers of social science such as Elder-Vass (2012) and Kaidesoja (2013). Where Elder-Vass's 'realist constructionist' ontology combines moderate constructionism and moderate realism, Kaidesoja's 'naturalized' realist social ontology assumes that social phenomena are a part of nature, and thus theories within a naturalist social ontology are expected to be compatible with the well-established assumptions and presuppositions of epistemically successful natural and physical sciences. Against transcendental and a priori epistemological and ontological reflection, Kaidesoja argues that just as for theories in the natural sciences, naturalist social ontology should be built 'by means of a posteriori arguments that take the epistemically successful scientific practices and well-confirmed results of different sciences as their premises. The relationship between theories developed in empirical sciences and in naturalist ontology should be seen as continuous' (2013, p. 203). Kaidesoja, much like Bourdieu (Lizardo, 2004), finds in cognitive science and related empirical fields resources that can be productively integrated into existing theoretical and methodological frameworks in the social sciences on the basis of his ontology.

Decoteau (2015) provides an illuminating overview of how Bourdieu's conception of social action and critical realism are, despite a number of important disagreements, compatible and complementary. Decoteau reworks Bourdieu's theory of habitus by suggesting that social selves are always situated at the intersection of multiple and competing social

locations (or field positions) and that the habitus itself is always layered. For Decoteau, reflexivity arises from disjunctures between field positions and across temporal sedimentation.

The compatibility of Bourdieusian theory with movements in the philosophy of social science suggests that he was, at a minimum, asking the right questions about the ontological and epistemological positions that are needed to undertake empirical sociological and, of special importance to Bourdieu and to contemporary sociology, interdisciplinary research.

Interdisciplinarity

The nature of digital technologies necessitates that digital social science continue to evolve as an interdisciplinary field. In parallel, Bourdieu's approach was inherently interdisciplinary, thus allowing his theoretical framework to engage with fields across the social sciences and serve as the foundation for digital social science. Just as scholars from across the social sciences study digital technologies, Bourdieu was well versed in theoretical approaches and concepts developed in disciplines other than sociology and organized a number of influential interdisciplinary projects. He developed the habitus concept from a study of the German art historian Erwin Panofsky, and the concept is daringly interdisciplinary in refusing to 'accept the institutionalized division of intellectual labor between contemporary psychology and sociology' (Swartz, 1997, p. 116; Lizardo, 2004). So while Bourdieu was influenced by art history, psychology, and cognitive science, he also worked closely with statisticians (Lebaron, 2009), and in 1975 launched the interdisciplinary journal *Actes de la Recherche en Sciences Sociales*. His interdisciplinarity not only allowed Bourdieu to draw from multiple schools of thought, but also ensured that his work would be used far beyond sociology. As we have previously discussed, his key strength is particularly apparent in the use of Bourdieusian concepts in interdisciplinary fields such as digital inequality that bring together scholars of sociology, communication, media studies, anthropology, etc.

Conclusions

The Bourdieusian framework has had a major impact on digital sociology and has provided valuable conceptual resources for what promises to be an increasingly important sociological subfield. A close analysis of studies of digital inequality sheds light on the degree to which Bourdieu's concepts have been applied for over 15 years to enrich the analysis of the social impacts of new digital technologies, while our review of the use of digital technologies to advance methodological innovation in the social sciences indicates the increasing importance of the concepts Bourdieu developed and of his overall relational approach to empirical research.

The rise of digitally mediated communication technology has created an entirely new realm for the application of the Bourdieusian framework, a realm which in many ways is tailor-made for concepts such as capital, field, and habitus. The affinity between the Bourdieusian framework and the digital realm compels us to ask why such concepts have risen to prominence within digital social science research. While we can offer no definitive answers to this question, we can propose a number of conjectures. First,

individuals' engagements with the digital realm throw into sharp relief new inter-relations between economic resources, internalized aptitudes, and social positioning. The diffusion of digital technology has created new pathways for the development of both primary and secondary forms of capital and habitus, as well as new fields of organized striving (Martin, 2003). One striking example, taken from a recent article on the automation of the oil drilling industry (Krauss, 2017), features a previously unemployed oil rig worker who has returned to steady employment, thanks to digital skill building, namely learning to operate joystick controls through his experience with video gaming.

But Bourdieu's framework has acquitted itself well in the digital age for more theoretical reasons as well. Bourdieu's ontological stance combining moderate realism and moderate social constructionism has proven to be a solid foundation for empirical sociology as well as for interdisciplinary learning and collaboration, and current developments in the philosophy of social science suggest that it was ahead of its time. Because his concepts emerged from and were intended to support empirical research practice, they have proved their value to the many researchers carrying out digital social science. Finally, the attention paid by Bourdieu to conceptual developments occurring in other disciplines, and his fruitful collaborations with statisticians and other researchers are in tune with the evolution of sociology into a relatively small but vibrantly interdisciplinary social science field (Moody & Light 2006).

Beyond recognizing the scope of Bourdieu's influence on digital sociology, our article suggests that digital sociologists who appropriate Bourdieu's ideas à la carte should recognize that field, capital, and habitus were developed together as part of an integrated relational approach to social research. Appropriating one of these concepts independent of the others allows for the needed flexibility in empirical research, but digital researchers who have found practical value in one or two parts of Bourdieu's sociology would do well to explore the rest of his oeuvre.

Disclosure statement

No potential conflict of interest was reported by the authors.

Notes on contributors

Gabe Ignatow is an Associate Professor of Sociology at the University of North Texas where he has taught since 2007. His research interests are in the areas of sociological theory, text mining and analysis methods, new media, and information policy. Gabe's current research involves working with computer scientists and statisticians to adapt text mining and topic modeling techniques for social science applications. He has served as the UNT Department of Sociology's graduate program co-director and undergraduate program director and has been selected as a faculty fellow at the Center for Cultural Sociology at Yale University. He is also a co-founder and the CEO of Grad-Trek, a graduate degree search engine company. Gabe has been working with mixed methods of text analysis since the 1990s, is the author of over 30 peer-reviewed articles and book chapters, and serves on the editorial boards of the journals *Sociological Forum*, *the Journal for the Theory of Social Behaviour*, and *Studies in Media and Communications*. His published books include: *An introduction to text mining* (with Rada Mihalcea, 2018) with Sage, *Text mining: A guidebook for the social sciences* (with Rada Mihalcea, 2017) with Sage, and *Transnational identity politics and the environment* (2007), with Lexington Books [email: ignatow@unt.edu].

Laura Robinson is Associate Professor in the Department of Sociology at Santa Clara University. She earned her PhD from UCLA, where she held a Mellon Fellowship in Latin American Studies and received a Bourse d'Accueil at the École Normale Supérieure. In addition to holding a postdoctoral fellowship on a John D. and Catherine T. MacArthur Foundation funded project at the USC Annenberg Center, Robinson has served as Affiliated Faculty at the ISSI at UC Berkeley, Visiting Assistant Professor at Cornell University, and Visiting Scholar at Trinity College Dublin. She is Series Co-Editor for Emerald Studies in Media and Communications and a past chair for the ASA Communication, Information Technologies, and Media Sociology section (CITAMS). Her publications include peer-reviewed articles in journals including *Sociological Methodology*, *Information, Communication & Society*, *New Media & Society*, and *Sociology*. Several of her publications have earned awards from CITASA, AOIR, and NCA IICD [email: laura@laurarobinson.org].

ORCID

Gabe Ignatow  <http://orcid.org/0000-0002-8091-6726>

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